

Solution Requirements

Date	15 July 2026
Team ID	xxxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviours, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviours (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	System allows farmers/users to securely register and log in using credentials	High
2	Authorization levels	Role-based access (Admin/User) to manage datasets, predictions, and reports	Medium

3	External interfaces	Integration with external services such as weather APIs and agricultural datasets for better prediction accuracy	Medium
4	Transactions processing	System processes soil and environmental data inputs, validates them, and sends them to the ML model for crop prediction.	High
5	Reporting	System generates crop recommendations along with insights, and displays results in a user-friendly format.	High
6	Business rules, etc.	Apply rules such as soil suitability thresholds, seasonal crop constraints, and environmental conditions before prediction	Medium
7	Compliance to laws or regulations	Ensure secure handling of user data and basic compliance with data protection and privacy standards	High
8	<i>Other</i>	Store prediction history and allow users to view past recommendations for analysis and improvement	Medium

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	System should quickly process input data and generate crop predictions without delay	Response time ≤ 2 seconds
2	Scalability	System should support increasing users and larger datasets efficiently	Supports up to 5000+ concurrent users
3	Security & Data Privacy	Ensure protection of user data through encryption and secure authentication mechanisms	HTTPS encryption, secure login system
4	Reliability & Availability	System should remain operational with minimal downtime and consistent performance	99.5% uptime
5	Usability & Accessibility	Provide an easy-to-use interface for farmers with minimal technical knowledge	Simple UI, mobile-friendly design
6	<i>Other</i>	System should be maintainable and portable across platforms for future enhancements	Modular design, supports web/mobile platforms